Homomorphic Encryption - The BFV Scheme

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Cryptography

With the expansion of global digitization, many parts of peoples lives are taking place in an online world. A persons Information is available all across the internet. With growing demand for online services a similarly growing demand in security and the uprising of cryptography was inevitable. Modern cryptography can be described as the "study of mathematical techniques for securing digital information, systems, and distributed computations against adversarial attacks."[1]

Homomorphic Encryption

In the past, operating on data required decrypting thus becoming vulnerable to attacks. Homomorphic Enryption allows for direct computations on the encrypted data. It follows the principles of a homomorphism and can broadly be written as follows:

$$\texttt{Dec}(\texttt{Enc}(x) \star \texttt{Enc}(y)) = x \star y$$

Its usecases span over several fields such as Cloud Computing, Machine Learning as well as the IoT. Performing statistical analyses or queries on sensitive data becomes possible without the loss of privacy. [2], [3]

Homomorphic Schemes

Partially Homomorphic Encryption

PHE allows unlimited operations of a single type - either addition or multiplication.

Somewhat Homomorphic Encryption

SHE supports both but is restricted to limited computational depths due to noise accumulation.

Fully Homomorphic Encryption

FHE resolves the limitations of SHE and PHE via bootstrapping, refreshing ciphertexts to evaluate arbitrary-depth circuits.

Capabilities

Within **FHE**, modern schemes are optimized for specific data types and applications. **BGV** and **BFV** are standards for exact modular arithmetic, leveraging Single Instruction, Multiple Data (SIMD) packing to process thousands of values per ciphertext. Conversely, **CKKS** is preferred for real or complex numbers—such as in privacy-preserving machine learning—due to its efficient noise management in approximate arithmetic. Finally, **TFHE** and **FHEW** excel at evaluating boolean circuits and non-linear functions through fast, programmable bootstrapping.

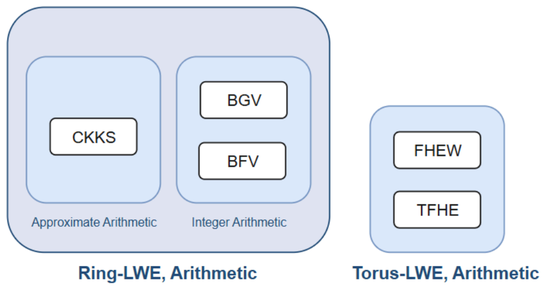


Figure 1: Categorization of the common schemes

The SH-BFV Scheme

Preliminaries

To construct the BFV Scheme as in [4], first define the most important variables. Security parameter λ defines the bit level of security. Consider $\mathbb{Z}[x]/(X^d+1)$ with $d = 2^n$. Let $q, t > 1$ resemble the cipher- and plaintext modulus. The scaling factor $\Delta = \lfloor \frac{q}{t} \rfloor$ and $\mathcal{R}_q$ is the set of polynomials with coefficients in $\mathbb{Z}_q = (-\frac{q}{2}, ..., \frac{q}{2}]$. Likewise for $\mathcal{R}_t$. Also assume a Discrete Gaussian distribution χ. Note that $\texttt{sk}$ will refer to secret key, $\texttt{pk}$ to public key and $\texttt{rlk}$ to the relinearization key.

Secret Key Generation

Take in λ, sample $\boldsymbol{s} \xleftarrow{\chi} \mathcal{R}_2$ and output $\texttt{sk} = \boldsymbol{s}$

Public Key Generation

Take in $\texttt{sk}$, sample $\boldsymbol{a} \xleftarrow{\chi} \mathcal{R}_q$ and $\boldsymbol{e} \longleftarrow \chi$ and output:

$$\texttt{pk} = \left([-(\boldsymbol{a} \cdot \boldsymbol{s} + \boldsymbol{e})]_q, \boldsymbol{a}\right)$$

Key Generation Evaluation V1

Assume $p = q^k$ and χ' as a Discrete Gaussian with a variance $\sigma' > \sigma$ from χ. Take $(\texttt{sk}, p)$, sample $\boldsymbol{a} \xleftarrow{\chi} \mathcal{R}_{p \cdot q}$, $\boldsymbol{e} \longleftarrow \chi'$ and return:

$$\texttt{rlk} = \left([-(\boldsymbol{a} \cdot \boldsymbol{s} + \boldsymbol{e}) + p \cdot \boldsymbol{s}^2]_{p \cdot q}, \boldsymbol{a}\right)$$

Encryption

Let $\boldsymbol{m} \in \mathcal{R}_t$, $\boldsymbol{p_0} = \texttt{pk}[0]$, $\boldsymbol{p_1} = \texttt{pk}[1]$, sample $\boldsymbol{u} \longleftarrow \mathcal{R}_2$, $\boldsymbol{e_1}, \boldsymbol{e_2} \longleftarrow \chi'$ and return:

$$\texttt{ct} = \left([\boldsymbol{p_0} \cdot \boldsymbol{u} + \boldsymbol{e_1} + \Delta \cdot \boldsymbol{m}]_q, [\boldsymbol{p_1} \cdot \boldsymbol{u} + \boldsymbol{e_2}]_q\right)$$

Decryption

Take $\texttt{sk}$ and $\texttt{ct}$ with $\boldsymbol{c_0} = \texttt{ct}[0]$ and $\boldsymbol{c_1} = \texttt{ct}[1]$, then compute:

$$m = \left[\left\lfloor \frac{t \cdot [\boldsymbol{c_0} + \boldsymbol{c_1} \cdot \boldsymbol{s}]_q}{q} \right\rceil\right]_t$$

Addition

Take $\texttt{ct}_1$, $\texttt{ct}_2$ and return:

$$\texttt{ct}_+ = \left([\texttt{ct}_1[0] + \texttt{ct}_2[0]]_q, [\texttt{ct}_1[1] + \texttt{ct}_2[1]]_q\right)$$

Multiplication

Take $\texttt{ct}_1$, $\texttt{ct}_2$, $\texttt{rlk}$ and compute:

$$\boldsymbol{c_0} = \left[\left\lfloor \left(\frac{t \cdot (\texttt{ct}_1[0] \cdot \texttt{ct}_2[0])}{q}\right) \right\rceil\right]_q$$

$$\boldsymbol{c_1} = \left[\left\lfloor \left(\frac{t \cdot (\texttt{ct}_1[0] \cdot \texttt{ct}_2[1] + \texttt{ct}_1[1] \cdot \texttt{ct}_2[0])}{q}\right) \right\rceil\right]_q$$

$$\boldsymbol{c_2} = \left[\left\lfloor \left(\frac{t \cdot (\texttt{ct}_1[1] \cdot \texttt{ct}_2[1])}{q}\right) \right\rceil\right]_q$$

To **Relinearize**, calculate:

$$(\boldsymbol{c_{2,0}}, \boldsymbol{c_{2,1}}) = \left(\left[\left\lfloor \frac{\boldsymbol{c_2} \cdot \texttt{rlk}[0]}{p} \right\rceil\right]_q, \left[\left\lfloor \frac{\boldsymbol{c_2} \cdot \texttt{rlk}[1]}{p} \right\rceil\right]_q\right)$$

Then return

$$\left([\boldsymbol{c_0} + \boldsymbol{c_{2,0}}]_q, [\boldsymbol{c_1} + \boldsymbol{c_{2,1}}]_q\right)$$

Applied SH-BFV

Preliminiaries

Take $\mathbb{Z}[X]/(X^2+1)$, consider $q = 120, t = 12$, thus $\Delta = 10$. Let $\boldsymbol{s} = x + 1$. Also let $\texttt{pk} = (\boldsymbol{p_0}, \boldsymbol{p_1})$ with $\boldsymbol{p_1} = \boldsymbol{a} = 10x$ and $\boldsymbol{e} = 1$. It follows from $\boldsymbol{p_0} = -(\boldsymbol{a} \cdot \boldsymbol{s} + \boldsymbol{e})$ that $\boldsymbol{p_0} = -(10x \cdot (x+1) + 1) = -10x + 9$. This concludes to:

$$\texttt{sk} = (x+1) \wedge \texttt{pk} = ([-10x + 9], 10x)$$

Encryption

Let $\boldsymbol{u} \longleftarrow \chi$. Consider $\boldsymbol{m_1} = 2, \boldsymbol{m_2} = x, \boldsymbol{u} = -x, \boldsymbol{e_1} = 1 - x$ and $\boldsymbol{e_2} = -1$.

$$\begin{aligned}\texttt{ct}_1[0] &= \underbrace{(-10x+9)}_{\boldsymbol{p_0}}\underbrace{(-x)}_{\boldsymbol{u}} + \underbrace{(1-x)}_{\boldsymbol{e_1}} + \underbrace{20}_{\Delta \boldsymbol{m}} \\ &= -10x + 11 \\ \texttt{ct}_1[1] &= \underbrace{(10x)}_{\boldsymbol{p_1}}\underbrace{(-x)}_{\boldsymbol{u}} + \underbrace{(-1)}_{\boldsymbol{e_2}} \\ &= 9\end{aligned}$$

It follows, that

$$\texttt{ct}_1 = ([-10x + 11], 9) = \texttt{Enc}(\boldsymbol{m_1})$$

Applying the same for $\boldsymbol{m_2}$ leads to

$$\texttt{ct}_2 = ([9x + 21], [11x - 11]) = \texttt{Enc}(\boldsymbol{m_2})$$

Multiplication

Applying the multiplication step from theory, $\texttt{ct}_1 \cdot \texttt{ct}_2$ results in tuple $\texttt{ct}_\times = (\boldsymbol{c_0}, \boldsymbol{c_1}, \boldsymbol{c_2})$ which in this example equals

$$\texttt{ct}_\times = ([-11x + 32], [31x + 18], [10x - 10])$$

Key Generation Evaluation

Let $p = 100, \boldsymbol{a} = 50 \in \mathcal{R}_{p \cdot q}$ and $\boldsymbol{e} = 0$. This yields

$$\texttt{rlk} = ([150x - 50], 50)$$

Relinearization

Applying given formula concludes the following relinearized ciphertext

$$\texttt{ct}_{\text{Re}} = ([-31x + 22], [36x + 13])$$

Decrytion

Plugging $\texttt{ct}_{\text{Re}}$ and $\texttt{sk}$ into the decryption formula will result in

$$\begin{aligned}\boldsymbol{m}_\times &= \left[\left\lfloor \frac{10 \cdot [(-31x + 22) + (36x + 13)(x+1)]_{100}}{100} \right\rceil\right]_{12} \\ &= 2x \\ &= \boldsymbol{m_1} \cdot \boldsymbol{m_2} \quad \square\end{aligned}$$

This examplifies the somewhat homomorphic property of the BFV Scheme.

References

[1] J. Katz and Y. Lindell, *Introduction to Modern Cryptography*. 2015.

[2] N. Aaraj, *Survey on Fully Homomorphic Encryption, Theory, and Applications*. 2022.

[3] P. Martins and others, *A Survey on Fully Homomorphic Encryption: An Engineering Perspective*. 2017.

[4] J. Fan and F. Vercauteren, *Somewhat Practical Fully Homomorphic Encryption*. 2012.